CERTAIN-COAST — illustrative figures

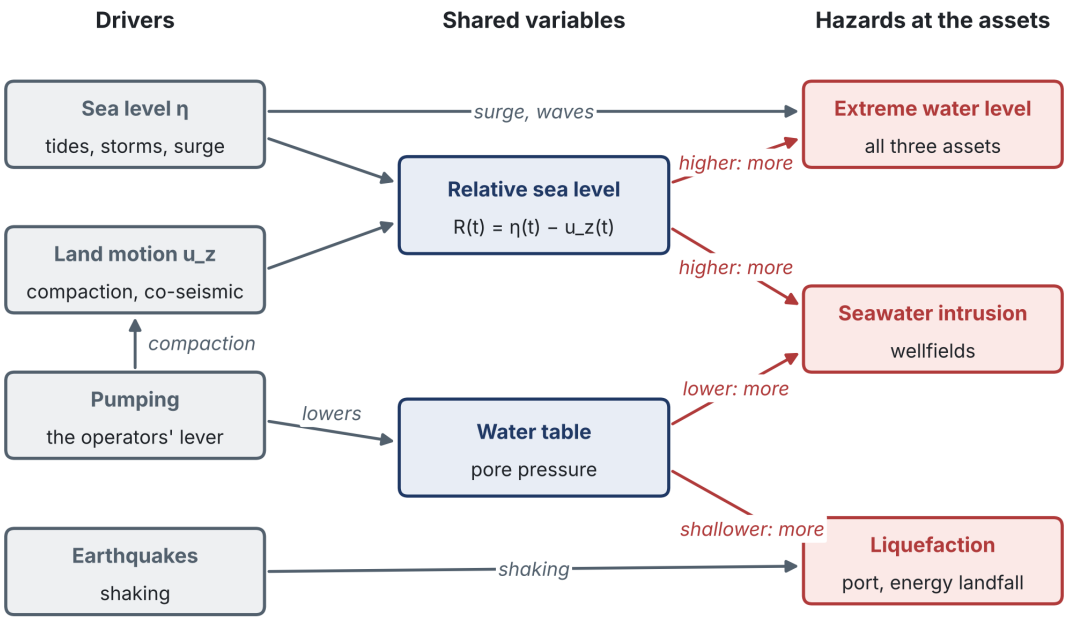
Certified early warning for seawater intrusion, liquefaction and extreme sea levels at the wellfields, port and energy landfall of a subsiding Adriatic coast

Optional upload of the "Project plan" tab of the SPS grant platform (NATO SPS call 2027-1). Twelve figures from the proposal documents and the preparatory reviews of 2026. Results of synthetic tests use illustrative values; they are not results at Durrës, which the project calibrates and verifies on site.

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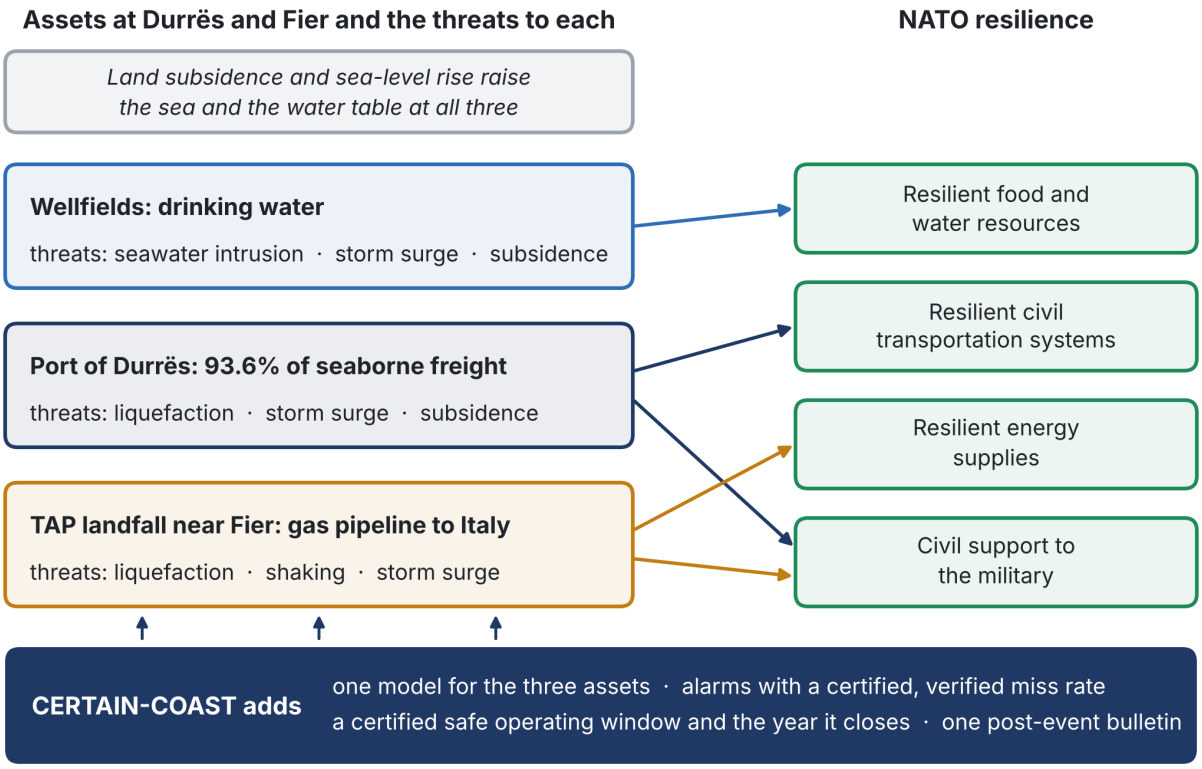
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Figure 1. The coast as one system



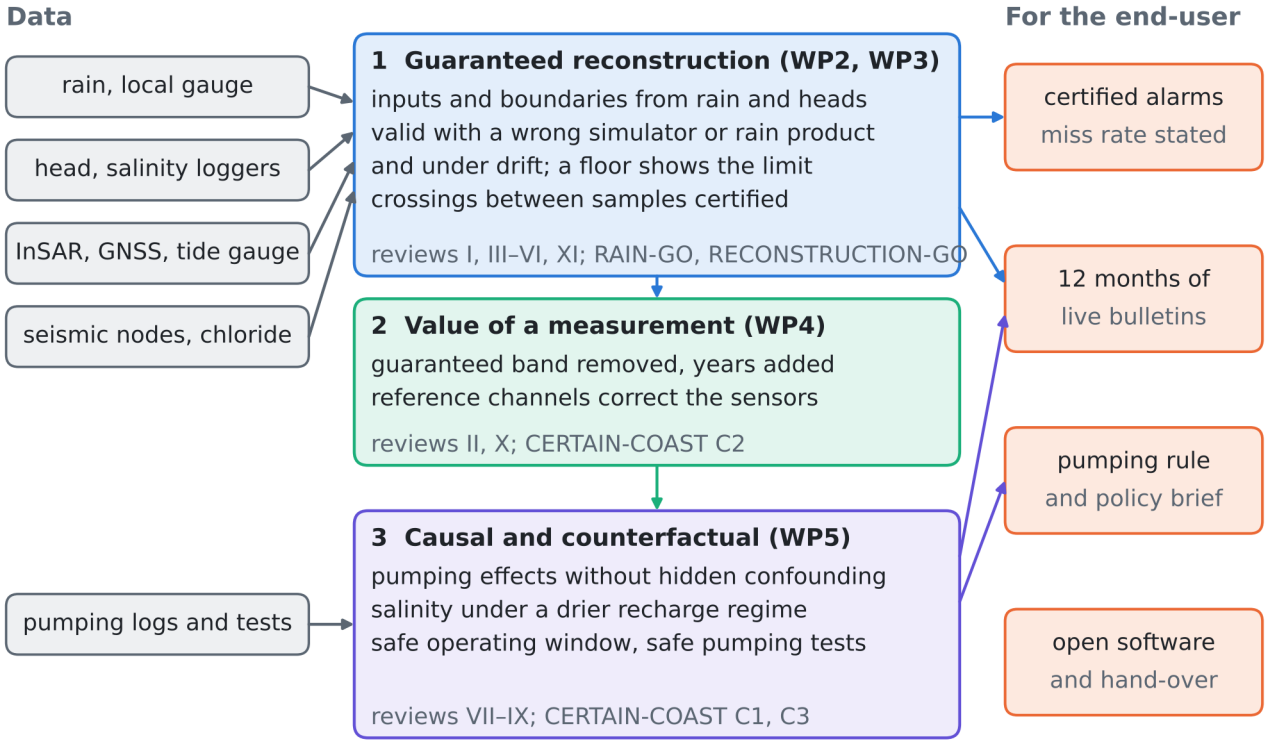
Four drivers act through two shared variables, relative sea level R(t) = η(t) − u_z(t) and the water table, on three hazards at the three assets. Red arrows: the direction in which each shared variable changes a hazard. Pumping, the operators' lever, lowers the water table: more intrusion and subsidence, less liquefaction.
Source: CERTAIN-COAST Introduction, Figure 1; State of the Art, Figure 1.

Figure 2. Critical assets, threats and NATO resilience



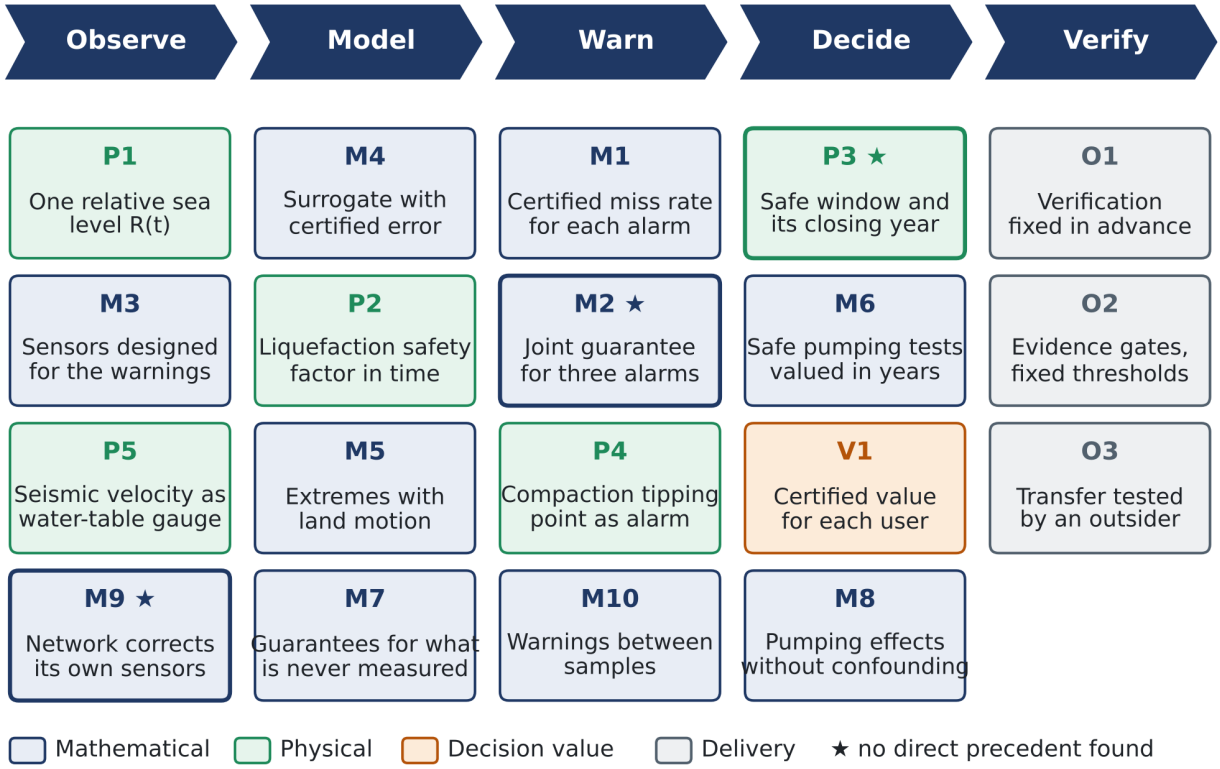
The threats to each asset at Durrës, the NATO baseline requirements for resilience that the assets serve, and what CERTAIN-COAST adds. Land subsidence and sea-level rise drive all three assets at once.
Source: CERTAIN-COAST Security Relevance, Figure 1.

Figure 3. The method in one picture



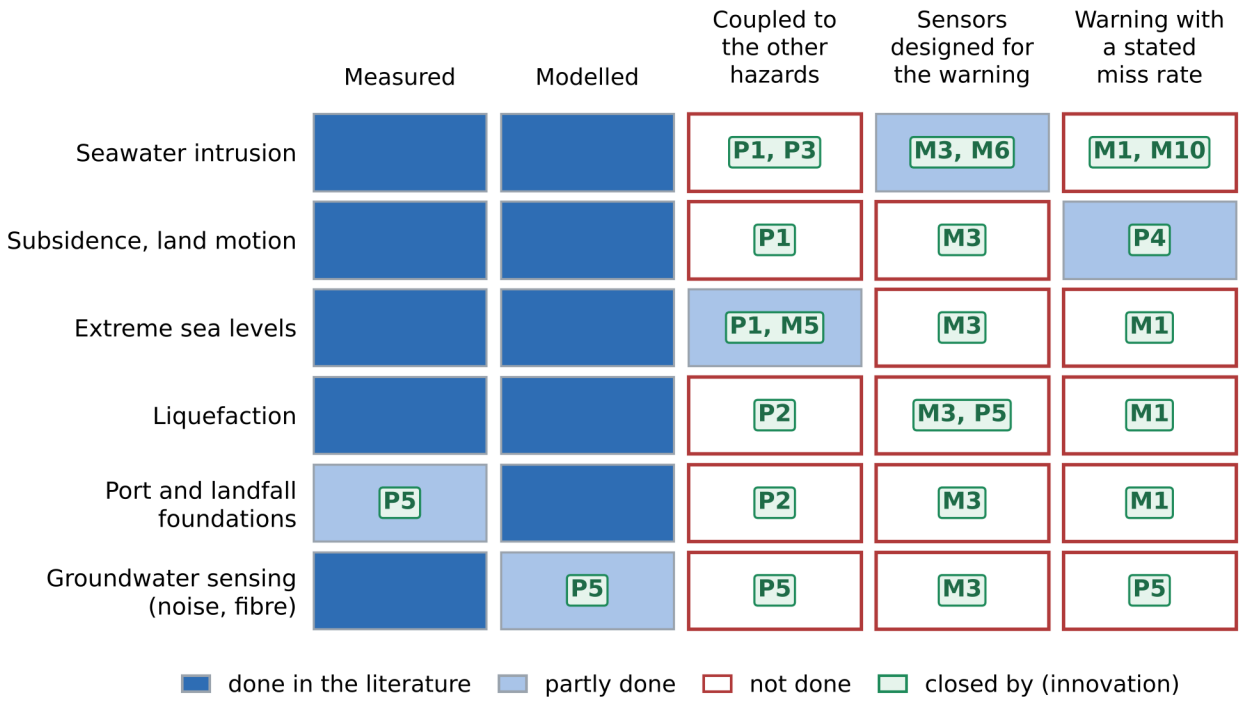
Data feed three pillars: (1) the inputs and boundaries nobody measures are reconstructed from rain and heads with guarantees that survive a wrong simulator, a wrong rain product and drift; (2) each sensor is valued by what it is guaranteed to buy before it is bought; (3) pumping effects, a drier climate and the safe operating window are computed with the hidden confounding removed and with stated error.
Source: CERTAIN-COAST Methodology for the NATO form, Figure 1.

Figure 4. Where each innovation acts



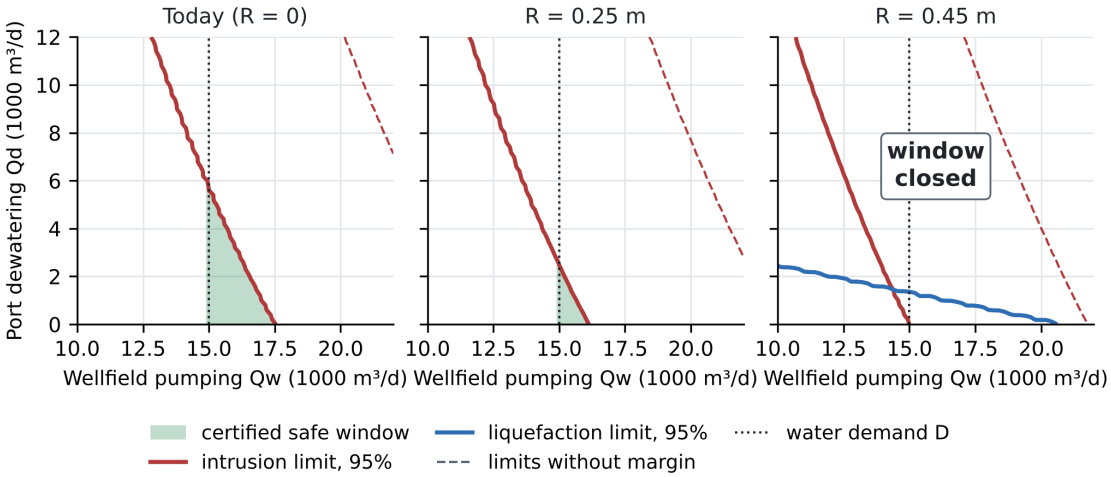
The sixteen technical innovations (M1–M10, P1–P5, V1) and three delivery innovations (O1–O3) placed in the warning chain, from observing the coast to verifying the warnings. ★: no direct precedent found in our search.
Source: CERTAIN-COAST Innovation, Figure 1.

Figure 5. The gaps the project closes



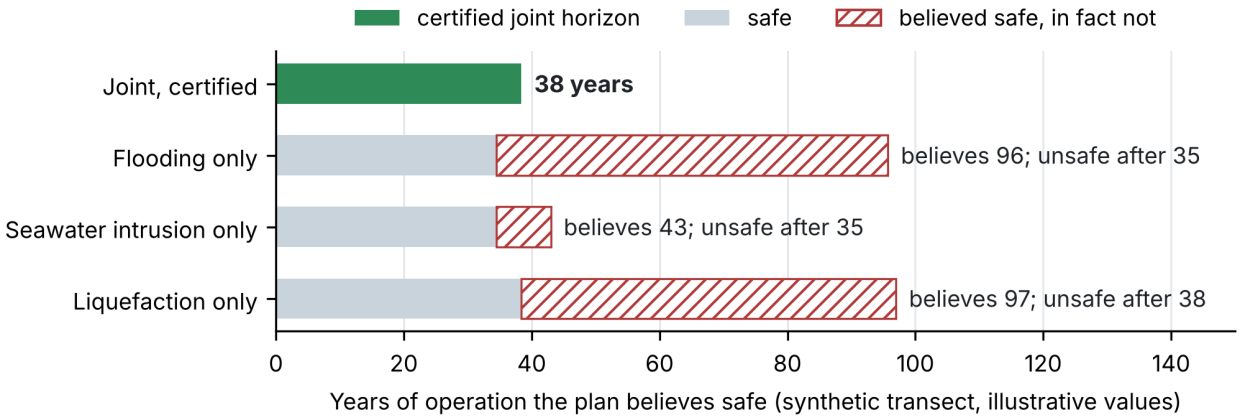
What the 2020–2026 literature has done for each component, and the CERTAIN-COAST innovation that closes each gap (codes as in the Innovation document).
Source: CERTAIN-COAST State of the Art, Figure 2.

Figure 6. The certified safe operating window



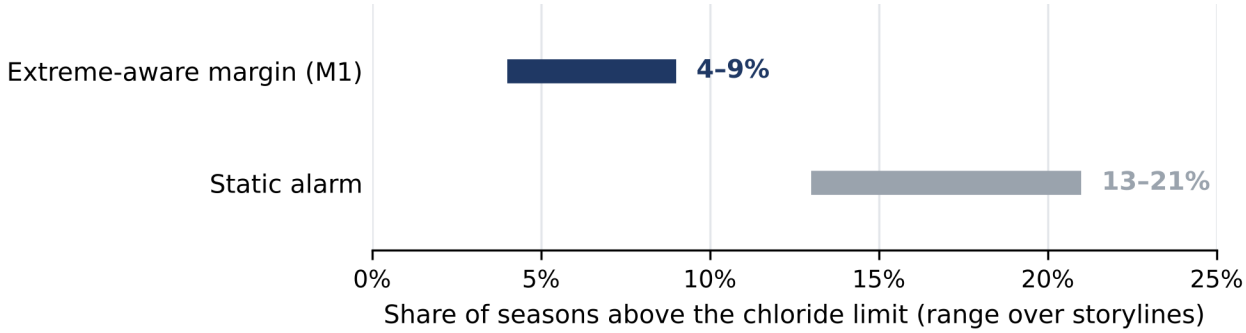
The certified safe window (green) in the plane of wellfield pumping and port dewatering, today and at two higher relative sea levels; it closes at R = 0.45 m. Red: the most dewatering before seawater reaches the wellfield (95%); blue: the least dewatering that keeps liquefaction within its limit (95%); dashed: the same limits without margins; dotted: the water demand. Synthetic transect with illustrative values.
Source: CERTAIN-COAST Methodology, Section 7.

Figure 7. Single-hazard plans believe they have decades more



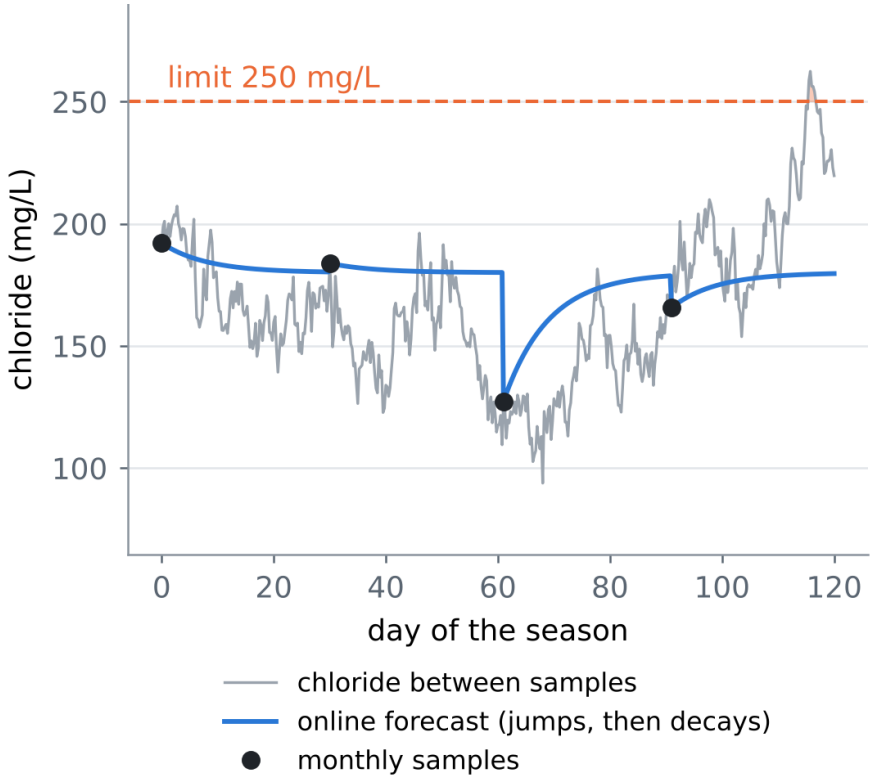
Years of safe operation believed by plans that watch one hazard, against the certified joint horizon of 38 years; hatched: years believed safe that are not. Synthetic transect with illustrative values.
Source: CERTAIN-COAST Security Relevance, Figure 2; Methodology, Section 7.3.

Figure 8. A certified alarm margin, tested before the project



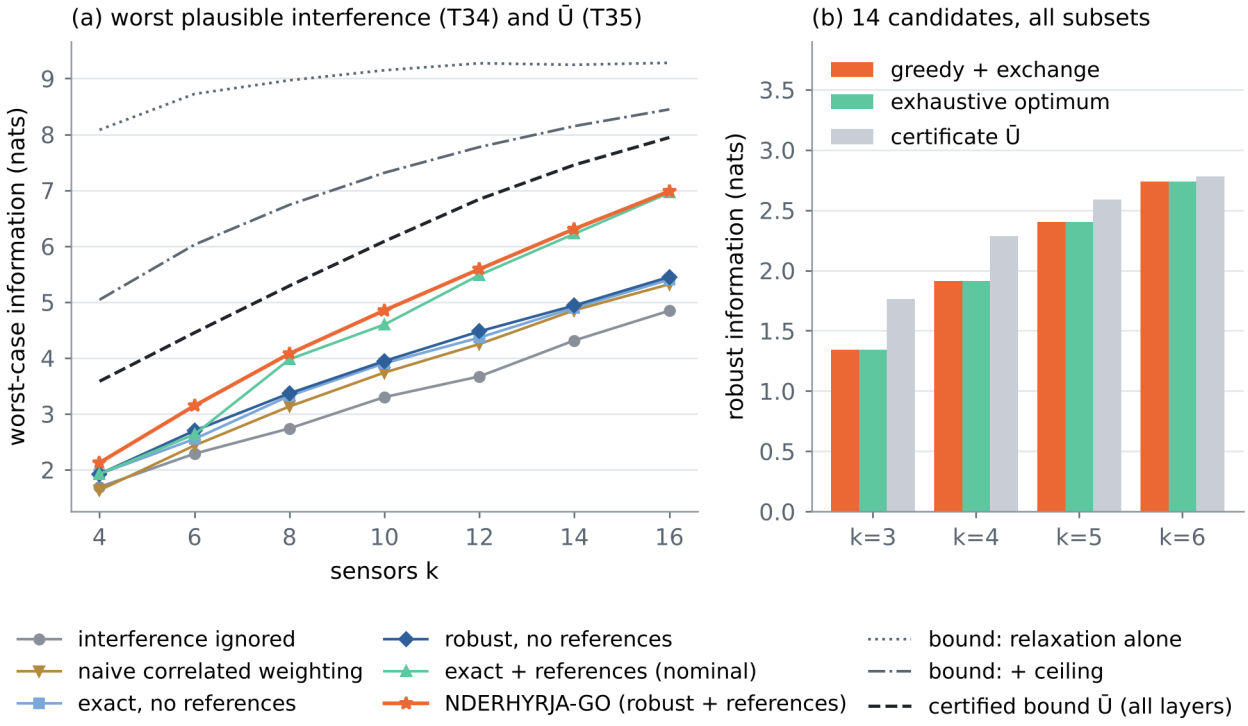
Range over storylines of the share of seasons above the chloride limit: 4–9% with the extreme-aware margin of M1, 13–21% with a static alarm. Tests on a synthetic coast, not results at Durrës.
Source: CERTAIN-COAST Innovation, Figure 2.

Figure 9. What happens between samples (review XI, PATH-GO)



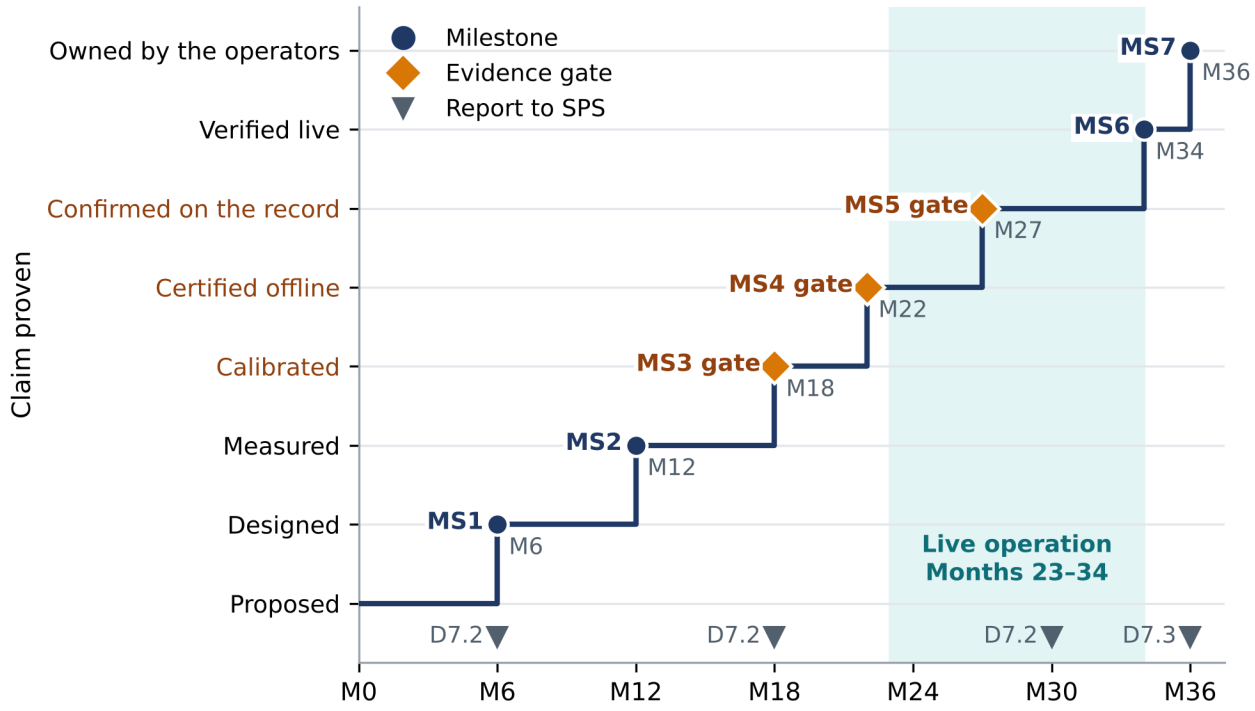
One simulated season with monthly samples: chloride crosses 250 mg/L between two samples that both look safe; the online forecast jumps at each sample and decays between them. In tests a crossing within the season was 34 times likelier than an exceedance on a sampling date; CERTAIN-COAST reports this chance as a certified band (M10).
Source: CERTAIN-COAST Review XI (PATH-GO), Figure 1, panel b.

Figure 10. A network that corrects its own sensors (review X, INTERVENTION-GO)



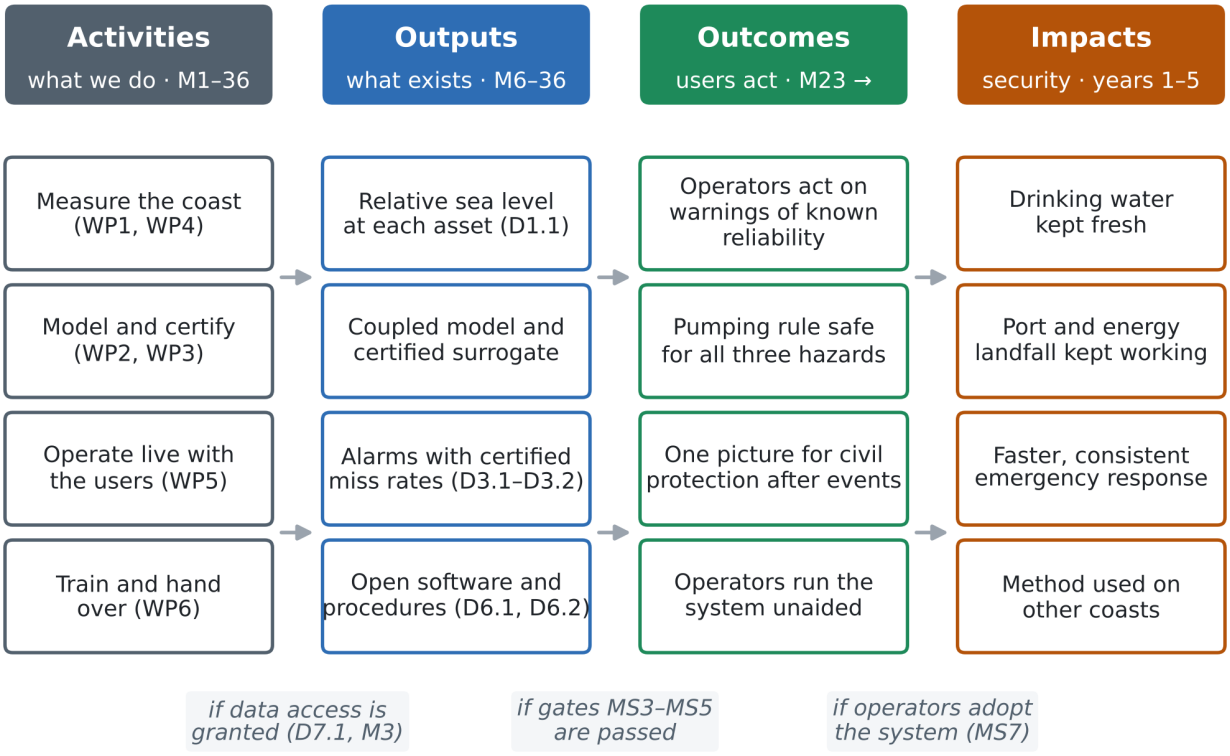
(a) Worst-case information of the network against the number of sensors when air pressure, logger drift, cable cross-talk and model error correlate the errors: buying a barometer and hub reference channels with the piezometers (orange) is best at every budget, and the certified bound Ū shows how far any network can be from the best one. (b) On 14 candidates the greedy-plus-exchange design equals the exhaustive optimum. Synthetic test bed (M9).
Source: CERTAIN-COAST Review X (INTERVENTION-GO), Figures T34–T35.

Figure 11. Milestones and evidence gates



Each milestone raises the claim the project can make by one rung; orange diamonds are the evidence gates (MS3, MS4, MS5) with thresholds fixed in advance; the shaded band is the twelve months of live operation; triangles mark the reports to the SPS Programme.
Source: CERTAIN-COAST Milestones, Figure 1.

Figure 12. Impact pathway



From the work packages to outputs (deliverables), outcomes (what the users do differently) and impacts. Italic notes: the condition for each step.
Source: CERTAIN-COAST Impact, Figure 1.